

Question Paper

Exam Date & Time: 25-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Cloud Computing [CSE_4063]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A large university, GlobalTech University, plans to implement a cloud-based Learning Management System (LMS) for students and faculty. The system will store student records, assignments, and research data. The university wants to: (5)
- A)

- Ensure data security and privacy for sensitive academic information,
- Allow scalability to handle thousands of users during exams or admissions,
- Facilitate collaboration with other universities for joint research projects, and
- Keep operational costs under control without maintaining too much hardware.

As a Cloud Computing Consultant, recommend the most appropriate cloud deployment model (Public, Private, Community, or Hybrid) for *GlobalTech University*.
Apply the features of the chosen model to justify how it meets the university's requirements.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) A small digital marketing company, *AdSphere Media*, is planning to launch a new web application that provides real-time campaign analytics for its clients. The company's goals are to: (5)

- Develop and deploy the application quickly without managing complex infrastructure,
- Use ready-made software tools for email marketing and data visualization,
- Scale resources automatically as the number of users increases, and
- Keep overall IT maintenance and cost low.

As a Cloud Computing Consultant, suggest the most suitable cloud service model(s) (*IaaS, PaaS, or SaaS*) for *AdSphere Media*.
Apply the features of the chosen model(s) to justify how they address the company's needs.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) A public cloud service provider needs to ensure data isolation while provisioning virtual machines to different tenants. (5)

- A)
- a) Explain how virtualization ensures logical isolation among multi-tenant environments.
 - b) Propose additional measures to strengthen VM security and tenant isolation.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) In a hybrid cloud, workloads are inconsistently balanced across VMs. (5)
- a) Analyze the resource allocation metrics that lead to imbalance.
 - b) Evaluate how dynamic VM migration can improve overall load distribution.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) A smart city uses a Message Bus with a Publish-Subscribe model to deliver traffic and pollution alerts to citizen apps. (5)
- A)
- a) Analyze how does the message bus reduce direct dependencies between services in this system?
 - b) Which filtering method (topic-based or content-based) would you apply for sending only pollution alerts to subscribed users? Justify.
 - c) Explain how event-driven alerts help city authorities take immediate action.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) A research lab runs a complex climate simulation workflow on a distributed grid system. The workflow includes sequential preprocessing tasks and parallel computation tasks that require frequent data sharing across nodes. (5)
- a) Analyze how a DAG-based workflow architecture ensures correct task execution and efficient resource usage for this simulation.
 - b) Explain how the workflow execution engine handles control flow vs. data flow and ensures fault tolerance during long-running scientific computations.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) A startup is developing a real-time food delivery tracking system. The application requires automatic scaling during peak meal times, fast deployment, and event-driven functions for processing live location updates. (5)
- A)
- a) Analyze how Google App Engine's Standard vs Flexible environments differ in supporting this application's scaling, runtime control, and execution requirements.
 - b) Evaluate whether AWS (Lambda, Elastic Beanstalk) or Microsoft Azure (Functions, App Services) provides a better programming environment for event-driven processing and rapid integration with backend storage - justify with service features.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) A cloud provider offers three kinds of agreements: one for individual users, one for enterprise customers, and one between internal departments of the same organization. Analyze your understanding of Service Level Agreements (SLAs) to identify the type of SLA represented in each case and explain how their scope and application differ in real-world cloud service scenarios. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Apply your knowledge of cloud security mechanisms to explain how encryption, strong authentication, and access control work together to protect sensitive customer data stored on a public cloud platform. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) A financial company uses a multi-layer cloud setup that includes firewalls, encryption, intrusion detection systems, and secure APIs to protect data across different service layers. (5)
- Explain how this setup reflects the concept of cloud computing security architecture and how layered protection enhances overall security.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	10

2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10

Total 50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4063:Cloud Computing

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	20(40%)
4	Analysis	6	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)